ECE3161 2024
Mini-Project
REPORT

TRUE-RMS-TO-DC CONVERTER

PROJECT TEAM ID

Team A1

PROJECT TEAM

Student ID #:	Student Full Name:
32521642	Benjamin Saunders
32481918	Jesse Thompson
32536437	Cullen Clifford

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Table of Contents

ACKNOWLEDGEMENT OF INTEGRITY.......... 3
Preface.......... 3
1. Introduction.......... 4
2. Literature Survey.......... 4
2.1 Over-arching Topology.......... 4
2.2 Key Principles.......... 4
2.2.1 The Explicit Method.......... 5
2.2.2 The Implicit Method.......... 5
2.3 Input adjustment.......... 5
2.4 Squaring / Square Rooting.......... 6
2.4.1 Explicit Square Rooting.......... 6
2.4.2 Log Anti-Log Squarer.......... 6
2.4.3 Translinear Current Squaring.......... 7
2.5 Averaging / Low Pass Filtering.......... 8
2.6 Final Topology.......... 8
3. Design Specifications.......... 8
3.1 Requirements Analysis.......... 8
3.2 Design Goals.......... 9
4. Design Description.......... 9
4.1. Theoretical Design.......... 9
4.1.1. Input rectification.......... 9
4.1.2 Voltage to Current Conversion.......... 10
4.1.3 Current Squarer Divider.......... 10
4.1.4 Low Pass filter.......... 11
4.1.5 Output adjustment.......... 12
4.2. Practical Considerations.......... 12
4.2.2 Component Tolerances.......... 12
4.2.3 Theoretical to real-world mapping.......... 12
4.3. Practical Design and Simulation.......... 13
4.3.1. Circuit Validation.......... 13
5. Physical Implementation.......... 15
5.1. Physical circuit annotations.......... 15
5.2 Component Variation and Tolerance.......... 15
5.2.1 MOSFET Variation.......... 15
6. Results and Discussion.......... 16
6.1 Discussion.......... 17
7. Conclusion.......... 17
7.1 Future Considerations.......... 18
REFERENCES.......... 19
APPENDIX.......... 21
Appendix A: Bill of Materials.......... 21
Appendix B: CMOS Squarer Calculations.......... 22
Appendix C: Low pass filter frequency response.......... 22
Appendix D: Circuit Validation Waveforms.......... 23
Appendix E: Waveform outputs.......... 25

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TEAM ID:

Student ID	Student Full Name	Signature
32521642	Benjamin Saunders	BenSaunders
32481918	Jesse Thompson	JesseThompson
32536437	Cullen Clifford	CullenClifford

PREFACE

To whom it may concern,

We fully understand that, at first glance, this document is very long and over-encumbered with figures, tables, and information. Hopefully, by the end, you will have noticed that all content and information has a purpose and shows the depths that we have gone to to get the most out of this project,

We take pride in what we have accomplished over the course of this project and believe that designing this. circuit from the ground up, and having it come out as a working product is an achievement in itself and has furthered our development as electrical engineers.

Good luck getting through this, and please be kind.

Cheers,

Team A1

1. Introduction

For this Project, we have developed an original design for an analog true-RMS-to-DC converter. The final design is an aggregation of our collective knowledge gained throughout ECE3161 and research into a multitude of circuit designs. The input waveform's peak-to-peak input voltage range is 100-200mV and has been tested to produce a result within the specifications from a DC input up to an AC signal of 10kHz for various waveforms. There are a multitude of real-world applications for this circuit, some of which include variable-frequency drive [1] in addition to low power and cost measurement equipment such as in a multimeter [2].

2. Literature Survey

True RMS (root-mean-square) conversion plays a critical role in the way we measure and analyse AC waves (both sinusoidal and non-sinusoidal). In mathematics, calculating "RMS," or the root-mean-square of a function, can be achieved with the following formula (Eq 2.1):

$$X_{RMS} = \sqrt{\frac{1}{T_0}\int_0^{T_0}[X(t)]^2 dt} \qquad \text{Eq 2.1}$$

2.1 Over-arching Topology

In an electrical context, the RMS value of an AC voltage is equal to the equivalent DC voltage that will carry the same amount of power. This is important for various systems, including variable-speed motor drives, computers, and HVAC. However, quickly and efficiently executing the calculation is not a trivial task for an analogue electrical circuit [3].

In a more simplified sense, the RMS voltage equation can be represented as follows:

$$V_{RMS} = \sqrt{\overline{|V_{in}|^2}} \qquad \text{Eq 2.2}$$

This must then be broken down into 5 main sections/operations [4]:

1. **Input Rectification:** Scaling and rectifying the input signal based on components 2 through 5 requirements.
2. **Squaring:** Squaring the signal as the internals of the integral in Eq 2.1. It is reliant on the rectification of the input waveform.
3. **Averaging:** Smoothing out the AC signal by taking the mean of the squared waveform. (Low pass filtering).
4. **Square-Rooting:** Reversing the non-linear gain from the squaring circuit.
5. **Output adjustment:** Accounting for the non-exact scale and offset accumulated through non-idealities in the circuit. Provided a high-impedance output as per the specifications.

2.2 Key Principles

Most of the key components of the above-described True RMS to DC converter can be trivially implemented through fundamental analogue electronics principles, with the non-linear squarer/square-rooter being the key exception. Extensive research into the area found two key topologies [5], one of which was recommended by the project brief [4], and both with their own benefits and drawbacks.

2.2.1 The Explicit Method

The explicit method (or direct method) is the most obvious method of computing a signal's RMS values. It is obtained by taking a linear approach to breaking down the RMS equation (Eq 2.2) and constructing blocks that achieve each operation. As seen below (Fig. 2.1), this method is the direct, naive implementation that performs each step in distinct and discrete stages.

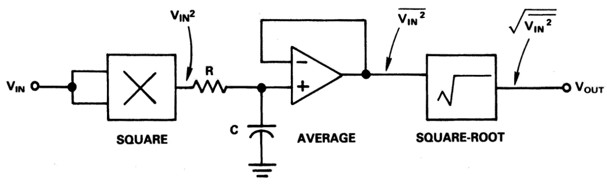
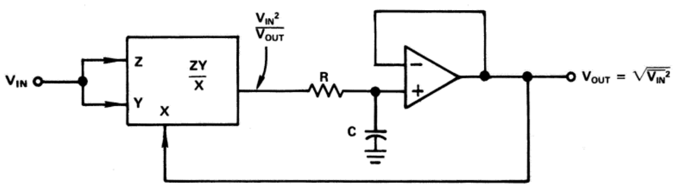


Figure 2.1: Explicit RMS computation method [5]

Whilst this method has no inherent bandwidth and speed problems compared to other designs, it comes with the fundamental problem of a limited dynamic range [5]. This means that for a relatively small input range, i.e., 50mV → 200mV, you must account for a much larger output range of 2.5V → 40V, greatly restricting the input's dynamic range and complicating the task when including the high voltage range involved with DC offsets.

2.2.2 The Implicit Method

The implicit method (or indirect method) utilizes feedback to aid with the square rooting portion of the converter [5]. Eq 2.3 shows how the RMS of an input voltage can be found by dividing the squared, averaged signal by the output RMS (Fig. 2.2). While this wouldn't work mathematically, the feedback from the system's output can be passed back through and used to aid its own processing [2][5].



$$V_{RMS} = \sqrt{\overline{(V_{in})^2}}$$

$$V_{RMS}^{\ 2} = \overline{(V_{in})^2}$$

$$V_{RMS} = \frac{\overline{(V_{in})^2}}{V_{RMS}} \qquad \text{Eq 2.3}$$

Figure 2.2: Implicit method RMS calculation.

This indirect approach removes the issue of a small dynamic range in the explicit method, with the added benefit of not needing an explicit square-rooting circuit (i.e. far less components). However, this method has a more restricted bandwidth than the explicit method because it relies on output feedback to complete the circuit. We found that the benefits of this method clearly outweigh the downsides, as supported by the current industry standards [2][6][7]. Going forward, we decided on using the implicit method, as making a decision this early provided clarity in the requirements going forward.

2.3 Input adjustment

Regardless of topology, rectifying the input waveform is the first step in all RMS circuits. A simple full-wave bridge rectifier was dismissed because our input voltages can fall below the ideal diode's forward voltage threshold (0.7V), causing inaccuracies for low-voltage inputs.

This led us to use a "Full-wave precision rectifier" design [8][9][10]. This rectifier leverages an op-amp's gain to keep diodes in their forward active region, ensuring accurate signal processing. While we considered several alternative precision rectifier designs [8][9][10] with better frequency response and small-signal performance, the simpler design adequately met our input specifications.

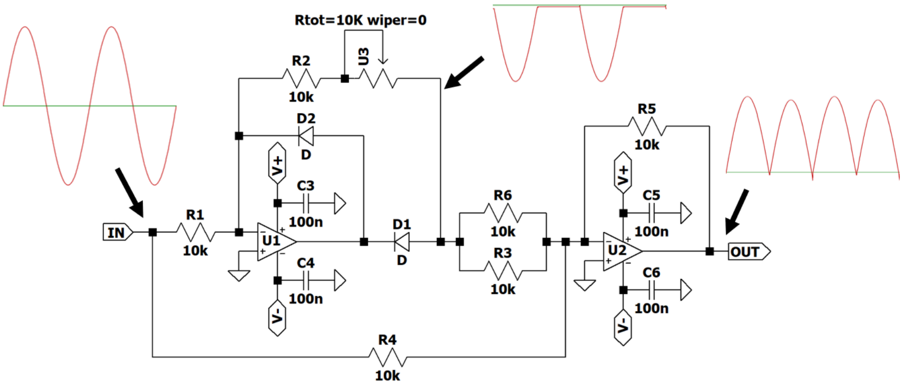


Figure 2.3: Full-wave precision rectifier, adapted from [7]

2.4 Squaring / Square Rooting

Applying non-linear operations to a signal is more complex than applying simple linear scales, offsets, and filters. Our research into squaring yielded another pair of common topologies, "Log Anti-log Squarer" and "Translinear current squaring."

You can then extract the square root of a signal using an op-amp with a squarer circuit in feedback [10]. This can be done with any of the aforementioned squaring designs, and is laid out below.

2.4.1 Explicit Square Rooting

We were taught that placing a nonlinear component in an opamp's feedback allows us to extract the inverse of that operation from its output [10].

We can deduce that placing a squaring element in the opamps feedback path can extract the input signal's root through this principle. This relationship is formed by the assumption that V- and V+ are equal on an ideal opamp, meaning that the output of the squarer is equal to Vin:

$$V_{in} = V_{out}^{2}$$

$$V_{out} = \sqrt{V_{in}} \qquad \text{Eq 2.4}$$

2.4.2 Log Anti-Log Squarer

By utilising non-linear components in an op-amps feedback path, it is possible to convert an input waveform into its relative logarithmic representation. These logarithms' multiplicative and quotient laws allow for direct mapping from simple, well-known op-amp operations like addition to complex non-linear operations like squaring [11].

$$2 \cdot log(V_{in}) \leftrightarrow log(V_{in}^{2}) \qquad \text{Eq 2.5}$$

This fundamental idea is then used to construct a log anti-log squaring circuit by converting the input to its logarithmic representation, doubling it, and then converting it back to its linear equivalent, giving the square of the input. A similar approach can be leveraged to achieve division in the implicit method converter design [5].

Whilst this design is fundamentally straightforward, implementation will likely contain a complex network of op-amps to facilitate the conversion to and from the log domain, limiting this circuit's operating frequency and maintaining a high power consumption.

2.4.3 Translinear Current Squaring

In our studies, we have learned that the current through a CMOS-FET in saturation has a non-linear gain [12] relative to the change in the FET's gate-source voltage.

$$I_d = \frac{K_m}{2} (V_{GS} - V_{th})^2 (1 + \lambda V_{ds}) \qquad \text{Eq 2.6}$$

This design is intended to operate with all of the FETs in the saturation region. As such we can assume that λ will be constant and simplify parts of the equation into the constant K out the front.

$$K = \frac{K_m}{2} (1 + \lambda V_{ds}) \qquad \text{Eq 2.7}$$

$$I_d = K(V_{GS} - V_{th})^2 \qquad \text{Eq 2.8}$$

This relation can then be used to create a circuit in which a combination of FETs performs a square and divide operation on its current input [13].

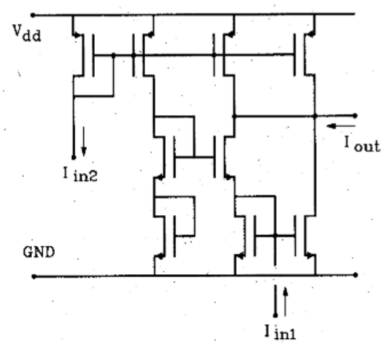


Figure 2.4: Theoretical Current Squarer Divider [13]

The operating equation for the above theoretical circuit can be modeled by:

$$I_{out} = \frac{I_{in1}^2}{8 I_{in2}} \qquad \text{Eq 2.9}$$

We can implement the implicit RMS conversion method when I_{in2} is sourced as feedback. It is important to realise that the circuit's reliance on Vth and Kn means that component matching will be needed to implement this design effectively.

The transistor-based nature of this circuit means that the overall power consumption of the circuit will be marginally lower than most designs since, as long as the FETs have the correct voltage biasing applied, the current can operate at very low magnitudes (~100s of nA). The theoretical model for this circuit is a much more compact method when compared to the many op-amps required to execute a log anti-log system [5][11]

2.5 Averaging / Low Pass Filtering

The rest of the converter's design required very little extra review outside of previous knowledge and the content taught in this unit. The averaging portion of the circuit is needed no matter what topology is chosen. However, depending on the methods chosen for squaring, the design must vary slightly.

The voltage and current squaring systems can use a passive low-pass filter to achieve averaging. For an active low-pass filter, however, our restriction to only voltage-comparator amps means that you would need to convert to and from the voltage domain before and after filtering.

The active low-pass filter has its benefits, though. The active components allow for an external DC voltage (+-5V rails) to boost the signal and provide a more stable output response. They also provide better rolloff characteristics and can perform better at lower frequencies. The main drawbacks of the active filter are its complexity, the need to convert between voltage and current, and the much greater power consumption than the passive filter's.

Consequently, we chose to use a passive filter, which can have a cut-off frequency in the desired range and offers significantly less complexity and power consumption than its alternative.

2.6 Final Topology

We used an implicit topology involving a precision rectifier, a MOSFET-based current squarer/divider circuit, a passive lowpass filter, and an output adjustment stage to account for any offsets and scaling applied throughout the circuit.

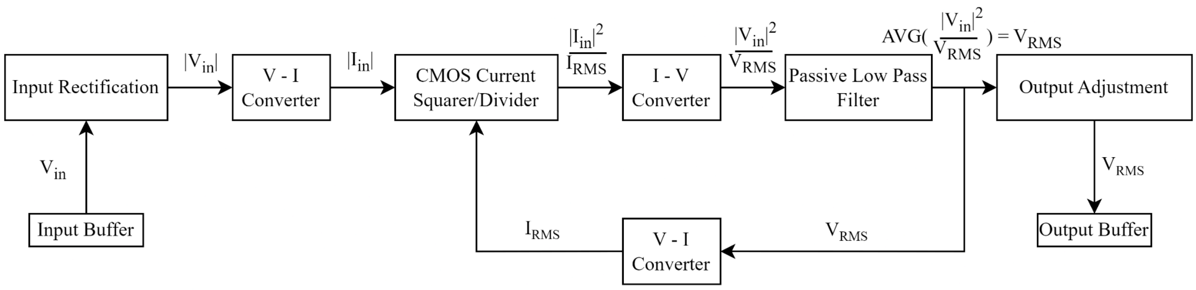


Figure 2.5: Theoretical circuit design based on literature review

3. Design Specifications

3.1 Requirements Analysis

The final true RMS to DC converter must meet a wide range of qualitative and quantitative operating criteria from the supplied project brief. Qualitatively, we are only permitted to use the components that were given to us as part of our component kit for the unit. This meant our design needed to account for tolerances, quantities, and substitute components if a specific component or the required amount was unavailable for a particular application.

All quantitative requirements are laid out in the table below:

Table 3.1: Supply Thresholds [4]

Specification	Minimum	Typical	Maximum
Power supply rails	-5V		5V
Supply current			200 mA
Power Consumption			1W

Table 3.2: Converter Criteria [4]

Specification	Minimum	Typical	Maximum
Peak-to-peak input voltage	100 mV		200 mV
Input frequency range	DC (1 Hz)		10 kHz
Voltage crest factor	1		3
Input Impedance	1 MΩ		
Input Capacitance			50 pF
Operation Temperature		25°C	
Output Error			3%
Output Impedance	1Ω		10 MΩ
Settling Time			10 s

All topological decisions were made with the above requirements (and design goals) in mind. We settled on the current-mode squaring circuit as its reliance on transistors allowed for much lower power consumption than that of the op-amp-reliant logarithmic circuit.

By buffering the input and output to the circuit, we can ensure we meet the input and output impedance requirements and minimize the input capacitance to the circuit.

3.2 Design Goals

From the beginning, it has been important that we meet all of the specified requirements provided above (Table. 3.1, 3.2) to ensure we make the most efficient converter possible. We were also incentivized to focus specifically on the following criteria to gain the most from this exercise.

1. DC Output error %
2. Power Consumption
3. Design Originality
4. Component Count

More specifically, our team strived to create a True RMS to DC converter that produced a remarkably accurate DC output while maintaining the lowest power consumption possible and still meeting all quantitative requirements.

Topology uniqueness was only specified as a recommended criterion, but our team decided that the final design must be fully designed and integrated by us, with the notable exception of the CMOS squarer-divider core [13]. This emphasis on originality came with the unfortunate side effect of a higher-than-average component count, but our team deemed this a suitable trade-off.

Note: We assume here that "component count" refers to discrete breadboard components; however, if an op-amp includes all components within, our design also provides a competitive component count.

4. Design Description

4.1. Theoretical Design

4.1.1. Input rectification

The precision rectifier consists of a half-wave rectifier and a summing amplifier. The half-wave rectifier rectifies the negative half of the wave but leaves the positive half. The summing amplifier then adds the input signal and the rectified half wave. This is done with scaling applied to the half-wave output by mismatching the two resistors for the input and half-wave such that the half-wave doubles its rectified amplitude. This means that when it is added to the input signal, it cancels out the negative part and adds itself, producing a fully rectified wave.

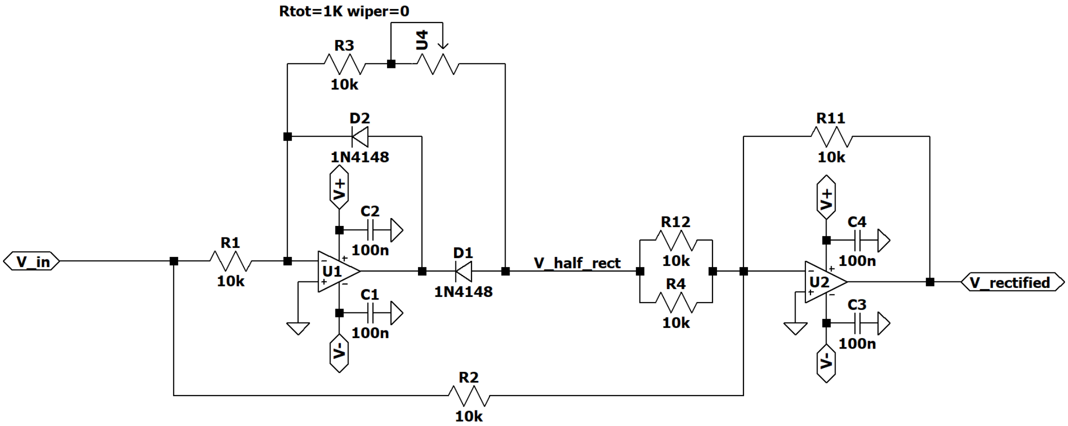


Figure 4.1: Theoretical Current Squarer [13]

Figure 4.1: LTSPICE implementation of the precision rectifier.

4.1.2 Voltage to Current Conversion

The linear relationship between current and voltage provided by Ohm's law means that a single resistor can accurately convert its voltage difference to current. The output current must then be mirrored to ensure a conversion that is independent of the load applied on the other end.

There were several considerations made when designing the mirror, including:

- Cascading an NPN and PNP mirror so that the current flow is maintained in the correct direction while avoiding the current offset resulting from a single PNP mirror in this orientation.
- Applying a negative 0.7 volt offset to the emitters of the NPN BJTs so that even when the input waveform is small (<700mV), the transistors will always have a large enough potential difference to turn on.

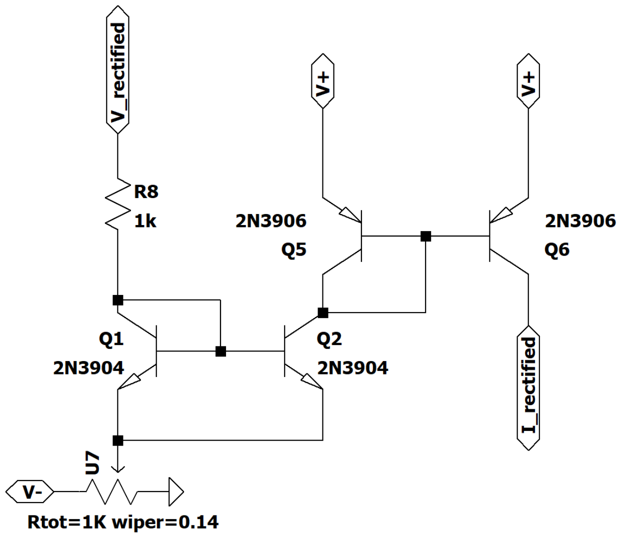


Figure 4.2: LTSPICE implementation of voltage to current converter

4.1.3 Current Squarer Divider.

The Current Squarer Divider we discovered in our research [13] can be divided into three parts: the squarer circuit, the biasing circuit, and the current mirrors.

The current squarer is governed by the following equation, which is

derived in Appendix

$$I_{out} = \frac{1}{2}K(V_2 - 2V_{th})^2 + \frac{I_{in}^{\ 2}}{2K(V_2 - 2V_{th})^2} \qquad \text{Eq 4.1}$$

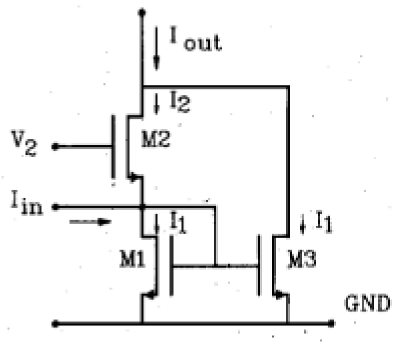


Figure 4.3: Theoretical Squarer circuit [13]

Next, we can add another FET-based circuit to map the biasing voltage (V_2) to an input current (Fig. 4.4). The function that governs the behavior of this circuit is as follows:

$$I_0 = \frac{1}{4}K(V_2 - 2V_{th})^2 \qquad \text{Eq 4.2}$$

If we connect the output V_2 of the biasing circuit in Figure 4.4 to the V_2 bias voltage input of the squaring circuit shown in Figure 4.3, then we can derive a much simpler equation for the overall circuit by substituting Eq 4.2 into Eq 4.1:

$$I_{out} = 2I_0 + \frac{I_{in}^2}{8I_0} \qquad \text{Eq 4.3}$$

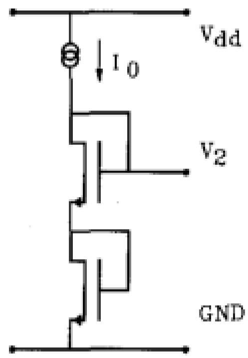


Figure 4.4: Theoretical Biasing circuit [13]

If we now treat I_0 as a second input, we can see in Eq 4.3 that we effectively already have a squarer divider circuit, but we have an extra offset that scales with I_0. We can account for this by introducing a triple-output current mirror and leveraging KCL to subtract 2 * I_0 from the output current. This gives the same final circuit shown in Figure 2.4 with the final equation (same as Eq 2.9):

$$I_{out} = \frac{I_{in1}^2}{8\,I_{in2}} \qquad \text{Eq 4.4}$$

It is also worth noting that in order for these equations to hold, we need to ensure that the FETs are always in saturation, leading to the following input restrictions:

$$\left|I_{in1}\right| < 4I_{in2} \qquad \text{Eq 4.5}$$

$$I_{in2} > 0 \qquad \text{Eq 4.6}$$

4.1.4 Low Pass filter.

We decided on a voltage-based passive low-pass filter for our circuit, which involved 3 sections to make it work. The output current from the squaring circuit must first be flipped before being converted into voltage over a 100kΩ resistor to ground. The resulting voltage can then be attenuated over a 470uF capacitor, resulting in a low-pass filter with a cut-off frequency measured at 80mHz (See Appendix C for frequency response).

After filtering, the voltage is then captured at V_rms_raw to propagate through a high-impedance channel to the output. All current flows through the 10k resistor (because of the high impedance of V_rms_raw), returning to a current-based signal in feedback to the squarer/divider.

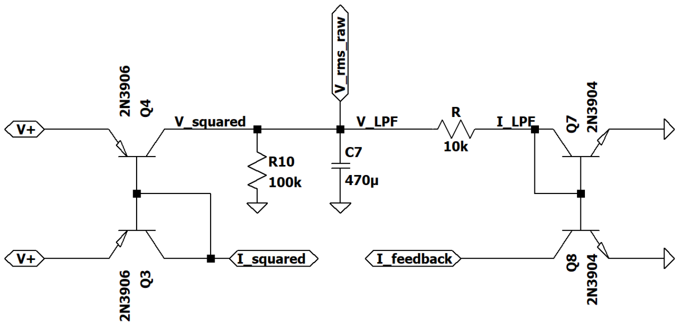


Figure 4.5: Low pass filter design

Disclaimer: After presenting our circuit, we realised, through hindsight, that placing a capacitor directly on the squarer output can low-pass the current and allow us to remove the V-I conversion altogether. See Section 7.1 for an implementation of this.

4.1.5 Output adjustment

Through non-idealities in the FETs and offsets incurred from current mirroring, we predicted the need for an output circuit to offset and scale the final RMS voltage by trimming each pot in the circuit below (Fig. 4.6). The fundamental design is a summing amplifier that offsets the input signal by some value between the +-5V rails controlled by U17. Then, it uses a potential divider with a potentiometer (U15) to attenuate the output and sufficiently match the True RMS values.

Finally, the output is then buffered with a TL084 [15] to meet the >1Ω output impedance requirements (the TL974 does not meet this requirement)

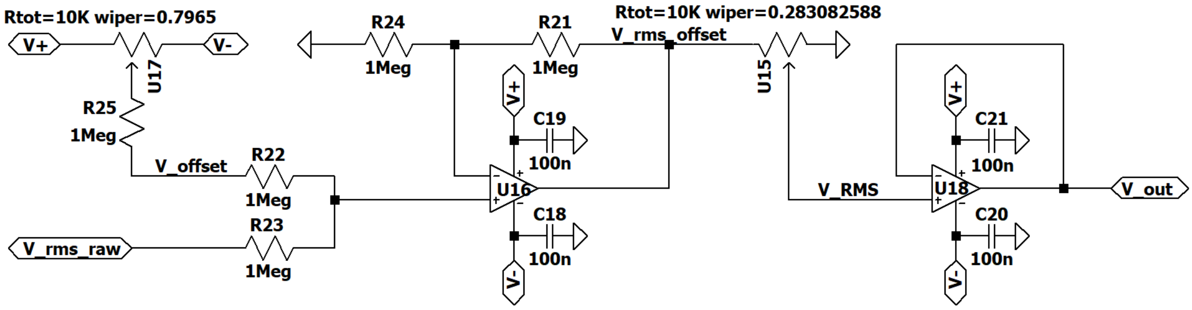


Figure 4.6: Output offset, scaling, and buffering

4.2. Practical Considerations

4.2.2 Component Tolerances

In the theoretical design of the precision rectifier, the resistance values were found to be critical to its accuracy. As discussed in [13], when selecting resistor values for use with an op-amp, their tolerance and magnitude need to be considered, with lower resistances enabling the best frequency response but the greatest power usage. The particular feedback configuration of the op-amp in the precision rectifier was susceptible to relatively small differences in resistance. To mitigate the effect on output error, a potentiometer was added to allow for tolerances to be corrected.

4.2.3 Theoretical to real-world mapping.

Due to component restrictions, the final design needed a single component substitution: the PMOS FETs in the circuit's current squarer/divider sub-section. In [13], they utilised three PMOS FETS to mirror current down each path, as shown in Figure 4.7; these were substituted with the 2N3906 PNP transistor provided in the kits. We could do this as PMOS FETS and PNP BJTs act similarly when used to mirror current. The only drawback of this substitution is the slight offset that they apply to the feedback path, but we can account for this in our output gain and offset circuit.

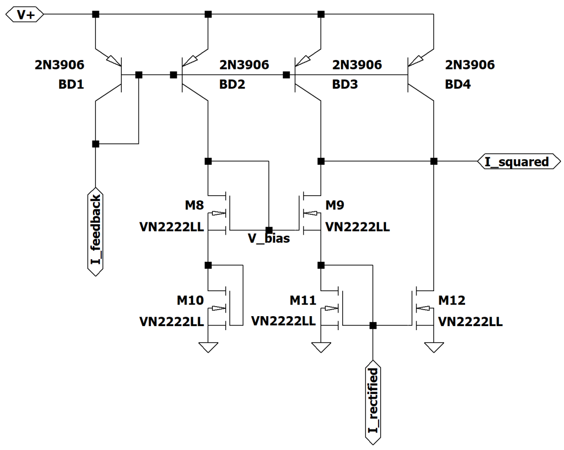


Figure 4.7: Adjusted Squaring circuit to remove PMOS transistors

4.3. Practical Design and Simulation

The schematic on the next page (Fig. 4.9) shows the entire circuit laid out in an LTSPICE schematic, including some extra practical considerations in addition to the ones specified above.

Due to non-idealities with the FETs in our kit, the output voltage was far greater than expected. Since the squarer has very delicate thresholds to meet (Eq. 4.5, 4.6), we were forced to swap out the output attenuation for an output gain instead of adjusting the input ranges.

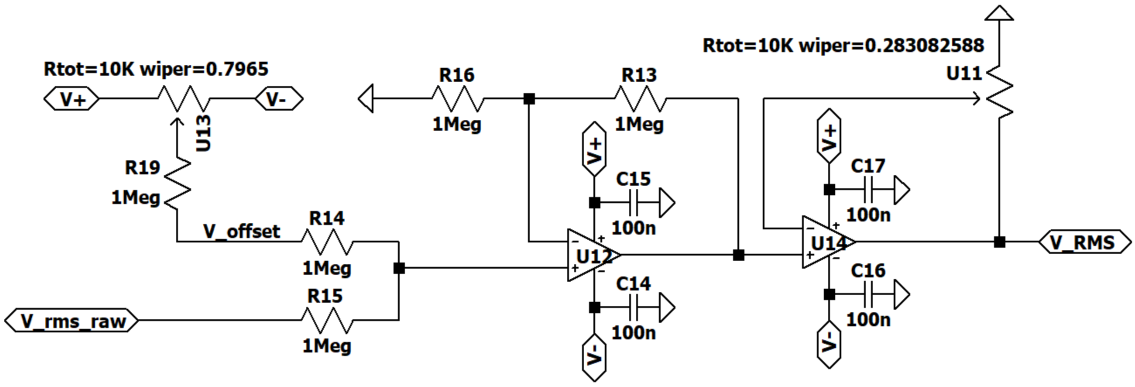


Figure 4.8: Practical output scaling and offset.

Now that we are gaining the output, any noise generated within the circuit was greatly exaggerated on the output, making it hard to get accurate readings. This led us to add an extra input scaling stage after rectification to lower the amount of gain at the end and, therefore, lower the effect of noise. The gain of this opamp was calculated based on the maximum input wave (1V DC + 0.2V AC) and the maximum output of the opamps (5V rails). This number reached a maximum theoretical gain of 4.16 and a practical gain of 3.4 to fit within E3 resistor values, accounting for a non-ideal power rail of <5V.

4.3.1. Circuit Validation

We validated each part of the practical circuit by probing the various nodes in the LTSPICE simulation. We programmed most of the specified input waveforms into a set of voltage sources that stepped through the input voltage range of 100mVpp to 200mVpp. Most notably, we measured and validated the following :

Note: for conciseness, all waveforms referenced below are taken for a 10kHz sine wave of 100mVpp, 150mVpp, and 200mVpp magnitudes

- The output of the precision rectifier (Appendix D.1)
- DC sweep of the squarer to ensure correct operation (Appendix D.2),
- Output of the low pass filter / Un-adjusted RMS value (Appendix D.3),
- Final circuit output (Appendix D.4)

A results table of the capabilities of this circuit can be found in Appendix D.5, where you will see an error margin of less than 3% for all but the PWM waveform. Speculation and discussion of these issues are laid out in Section 6 of this report.

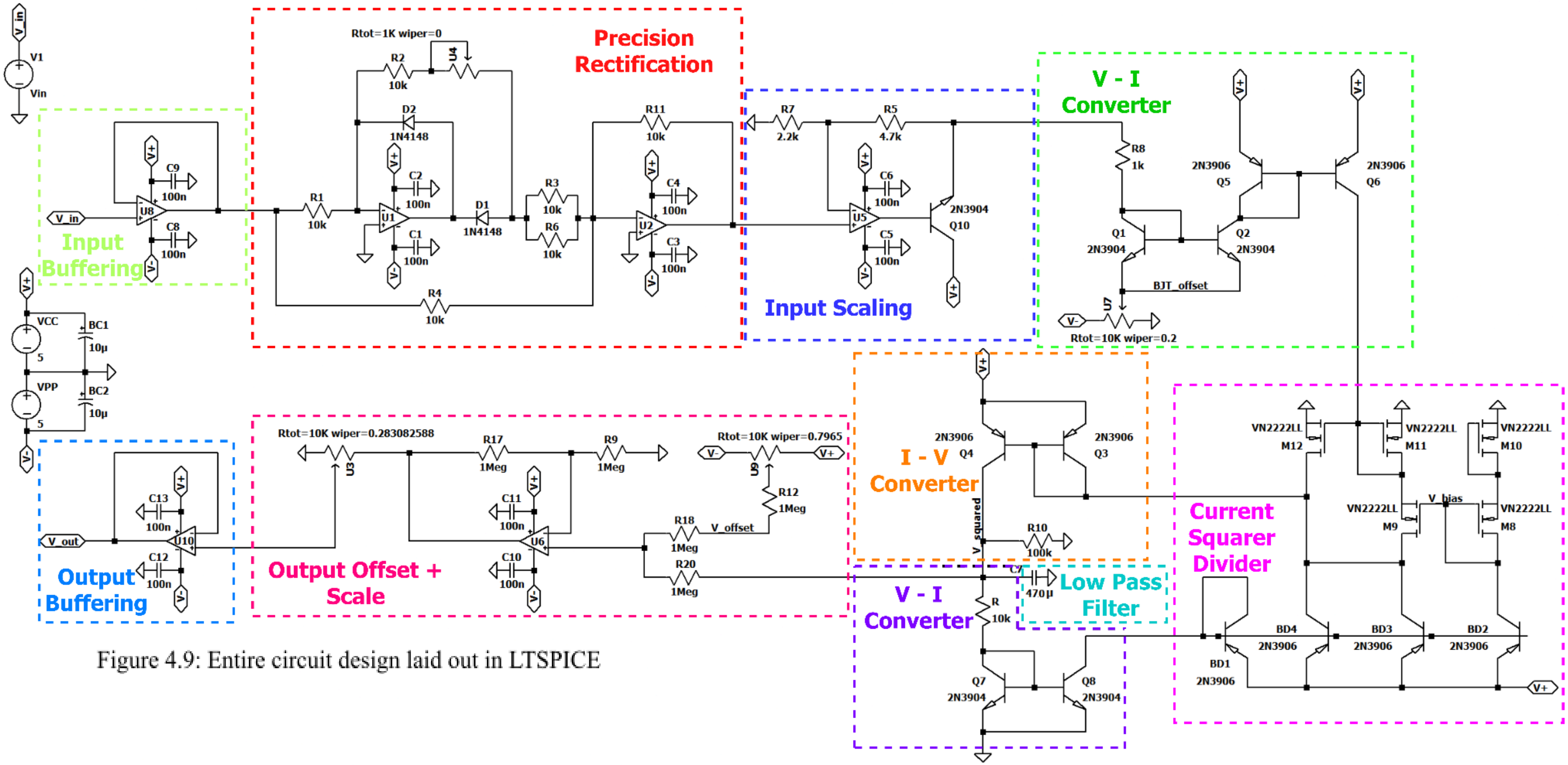


Figure 4.9: Entire circuit design laid out in LTSPICE

5. Physical Implementation

5.1. Physical circuit annotations

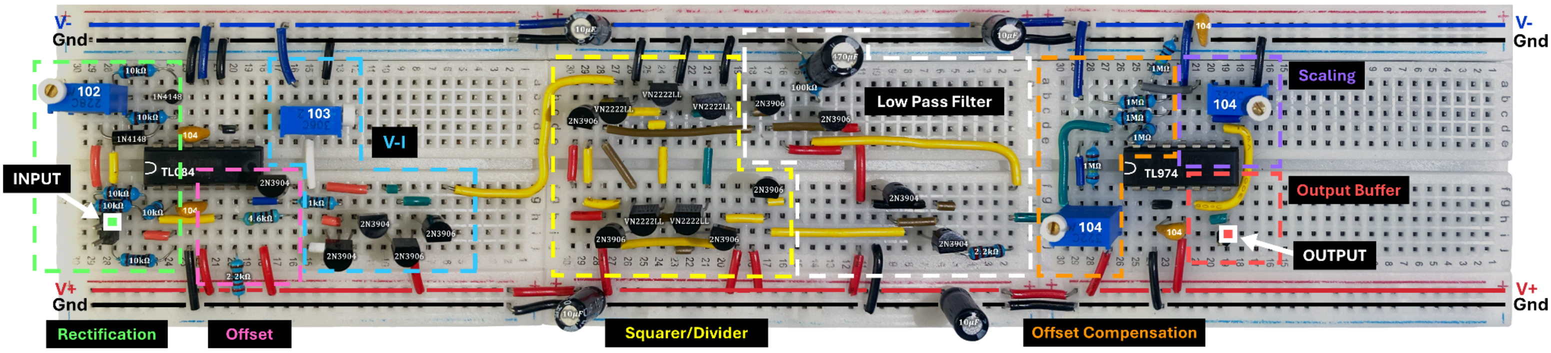


Figure 5.1 Annotated diagram of the circuits physical implementation

5.2 Component Variation and Tolerance

5.2.1 MOSFET Variation

Many of the steps taken in deriving the equation for the squarer divider circuit, such as from Eq 4.1 to Eq 4.2, make their simplifications assuming that all of the FETs used have the same V_{th}. Unfortunately, This is not always true for real FETs, particularly for discrete ones. Looking at the datasheet for the VN2222LL [14], V_{th} could be anywhere between 0.6 and 2.5 volts, which is significant and will affect the response of the circuit greatly.

To overcome this limitation, we decided to bin the FETs that we had available in order to keep their V_{th} values as close as possible. This was done by measuring each FET's I_D vs V_{DS} curve for a value of V_{GS}. From this, we recorded the corresponding I_D value and used that to compare it against the other FETs to determine which ones behave most similarly with respect to their I_D vs V_{GS} relations.

The binning process also minimized the very low-frequency (suspected) thermal noise we were experiencing. By matching the FETs, the output variation went from **+-50mV to +-3mV.**

5.2.2 Resistor Variation

When measuring the resistance of the resistors labeled as $10k\Omega$ in the provided components, there was a large variation, with some varying from the stated value by more than 10%. The $10k\Omega$ resistors used in the physical implementation were binned for those which were closest in value to each other. Additionally, during a calibration phase, all the potentiometers of each sub-section of the circuit were individually adjusted while viewing the output waveform.

All circuit validation and operation can be found below (Section 6: Results and Discussions)

6. Results and Discussion

The final circuit implementation was more herculean than we had predicted. As mentioned in the practical considerations, a few key changes were made to the circuit (output scaling instead of attenuating and input scaling) due to a crippling amount of noise.

We also faced an issue that arose due to chaining together so many transistors in the current squarer divider [13] Fig 4.7. Between the 5V rail and ground, we had a maximum chain of 3 transistors, each with their own voltage drops of between 0.6 and 2.5V [14], meaning that not having sufficient rail voltages was a very probable issue. Binning our FETs showed that we were provided a pretty linear distribution of Vds values between 0.6 and 2.5V, and selectively choosing which FETs we used allowed us to meet these threshold requirements with the added benefit of reducing (what we suspected to be) thermal noise (see Section 5.2.1).

Table 6.1: RMS Measurement Results (Appendix E)

Wave Type	Function (V)	Amplitude (mV)	Frequency (Hz)	Ideal RMS Value (mV)	Measured Value (mV)	% error
DC	1	100	0	100.0000	104.56	4.56
Sine Wave	$sin\ (t)$	100	2	70.7107	71.851	1.61
		200	2	141.4214	141.18	-0.17
		200	10k	141.4214	139.14	-1.61
Full Rectified Sine Wave	$\|sin\ (t)\|$	200	10k	141.4214	139.38	-1.44
Half Rectified Sine Wave	$\sum_{n=-\infty}^{\infty} HS(t + 2\pi n)$ *where* $HS(t) = \{ sin(t)\ if\ 0 \leq t \leq \pi$	200	10k	100.0000	76.240	-23.76
Triangle Wave	$\sum_{n=-\infty}^{\infty} \Lambda(t + n)$	100	10k	57.7350	58.069	0.58
Square Wave	$\sum_{n=-\infty}^{\infty} \Pi(t + n)$	100	10k	100.0000	101.65	1.65
PWM Signal*	$\sum_{n=-\infty}^{\infty} PWM(t + nT)$ *where* $PWM(t) = \{1\ if\ 0 \leq t \leq t_r\ 0$	200	10k	63.2456	24.879	-60.66

*PWM signal was measured at a duty cycle of 10%

6.1 Discussion

The above results table shows how our final implementation did a qualitatively decent job at converting true RMS to DC. Much like the simulation results (Appendix D.5), the circuit had a <3% DC output error for most input waveforms, except for the half-wave rectified sinusoid and the low-duty-cycle PWM signal. It is still somewhat unclear what causes these specific inputs to have a high output error, but through experimentation, we have deduced that AC and DC signals are being scaled differently, even though the frequency response of the LPF seems accurate. All waveform outputs can be found in Appendix E.

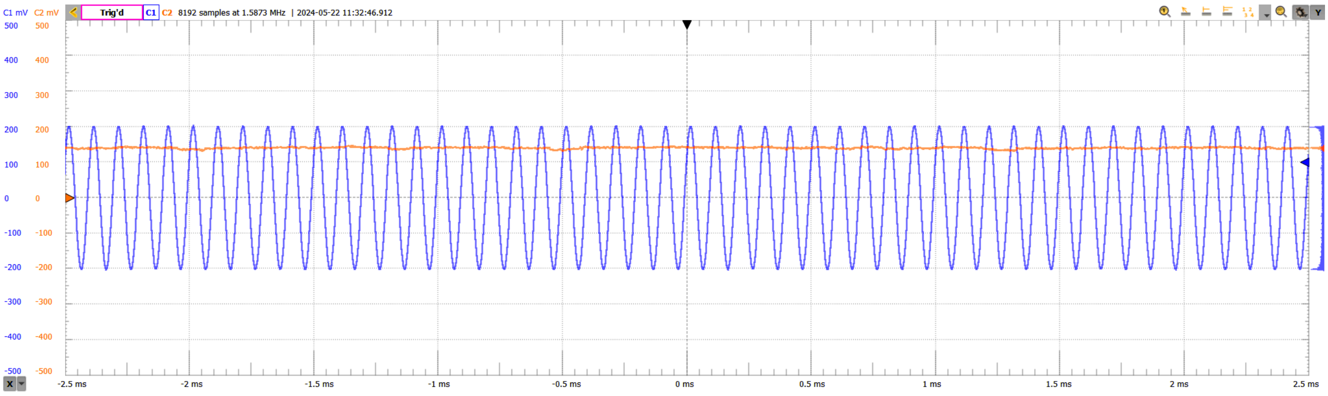


Figure 6,1: 100mV 10kHz sine input/output waveform

The above waveform (Fig 6.1) shows that the output contains some noise, but the +- 5mV oscillations are far better than the pre-binning +-50mV. (unfortunately, we don't have any images)

Even though comprehensive testing was not completed for all frequencies on all waveforms, it is apparent through the different sine-wave inputs that a change in input frequency has a negligible effect on the output voltage.

An interesting side note is that our circuit's operating current seems so low that having a phone nearby can cause some very serious RF interference.

7. Conclusion

Completing this project enabled us to achieve two main objectives: We researched, designed, validated, and tested a working True RMS-to-DC converter and gained a significant understanding and appreciation of the in-depth realm of real-world circuit design.

Through a deep dive into the world of non-ideal components and behaviour, we were able to utilise several methods to help reduce noise and improve the overall performance of our converter. In the end, our design closely matched most of the given requirements. Most importantly, our DC error percentage fell under 3% for almost all of the required input waves, except for the half-wave rectified wave and the low percentage PWM.

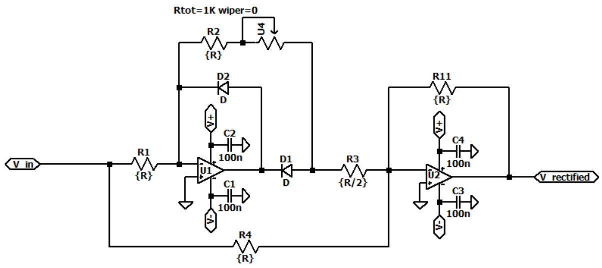
Our circuit's input impedance was well above 1MΩ and within 50pF capacitance, and we used a TL084 to achieve an output impedance within the >1Ω requirement.

Our final design had a simulated current draw of 8mA on the negative rail and 10mA on the positive rail, giving a total power consumption of 90mW for the whole circuit, which is significantly lower than the 1w requirement.

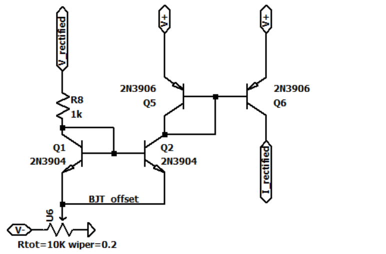
7.1 Future Considerations

After completing our circuit demonstration, we realized that our design could be significantly simplified and its performance improved. This is done by low-pass filtering in the current domain instead of the voltage domain, thereby eliminating two current mirrors from the feedback path and allowing for output voltage gain adjustment without the need for an amplifier.

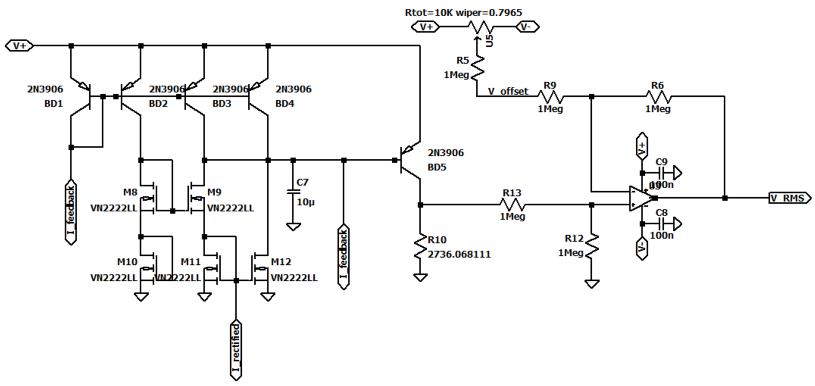
According to the simulation, the updated changes to the design would result in a current draw of 7.5mA on both rails and a total power consumption of 75mW, which is a 17% reduction in power consumption over our original design.



Precision Rectification



Voltage to Current Conversion



Current Squarer Divider

If you have made it this far, hopefully, you have been able to appreciate the work we put into this project and that we have been able to display the passion and effort that went into designing this complex circuit.

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APPENDIX

Include supplementary information here for assessment.

Appendix A: Bill of Materials

Provide a list of all components used in the True RMS to DC Converter, including part numbers and quantities

Part Name	Model	Value	Tolerance	Quantity
Op-amp	TL974	-	-	1
Op-amp	TL084	-	-	1
Resistors				
	5 band ¼ Watt	$1M\Omega$	1%	5
	5 band ¼ Watt	$100k\Omega$	1%	1
	5 band ¼ Watt	$10k\Omega$	1%	6
	5 band ¼ Watt	$4.6k\Omega$	1%	1
	5 band ¼ Watt	$2.2k\Omega$	1%	2
	5 band ¼ Watt	$1k\Omega$	1%	1
Potentiometers				
	W104	$100k\Omega$	-	2
	W103	$10k\Omega$	-	1
	W102	$1k\Omega$	-	1
Capacitors				
	Electrolytic	470μF	-	1
	Electrolytic	$10\mu F$	-	4
	Ceramic 104	$100nF$	-	4
NPN BJT	2N3904	-	-	5
PNP BJT	2N3906	-	-	8
NMOS	VN2222LL	-	-	5
			Total Number of Components	49

Appendix B: CMOS Squarer Calculations.

Firstly, the current squarer consists of three Transistors:

Looking first at M1 and M3, we can see that they share a common gate voltage and will act as a current mirror with the current being the sum of I_{in} and I_2, giving:

$$I_1 = I_{in} + I_2 \qquad \text{Eq B.1}$$

Using Eq 2.8 from earlier and looking at M2 and M3 we can derive an equation for I_1-I_2 and I_1+I_2:

$$I_1 - I_2 = K(V_2 - 2V_{th})(V_a - V_b) \qquad \text{Eq B.2}$$

$$I_1 + I_2 = \frac{1}{2}K(V_2 - 2V_{th})^2 + \frac{(I_1 - I_2)^2}{2K(V_2 - 2V_{th})^2} \qquad \text{Eq B.3}$$

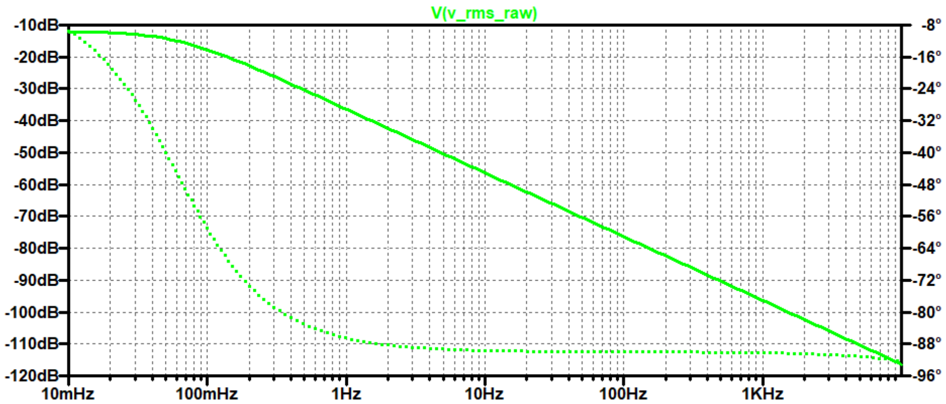
and finally we can also see:

$$I_{out} = I_1 + I_2 \qquad \text{Eq B.4}$$

Using these we can derive the equation for the squarer as:

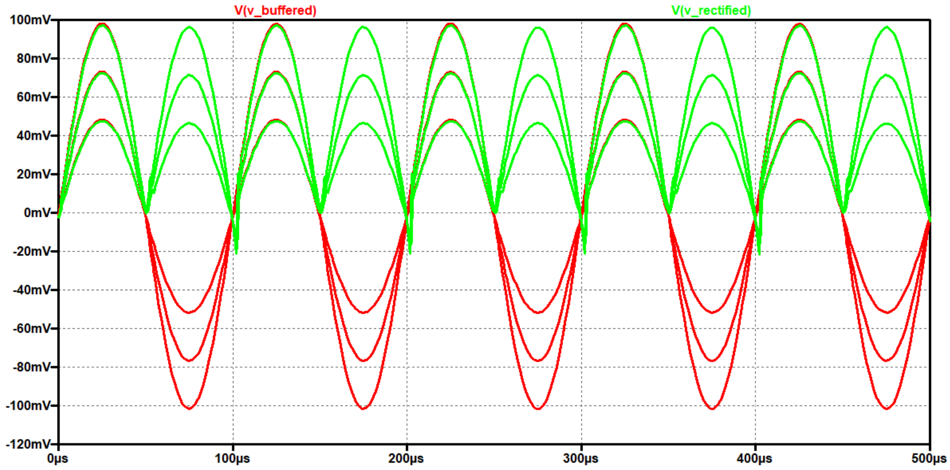
$$I_{out} = \frac{1}{2}K(V_2 - 2V_{th})^2 + \frac{I_{in}^2}{2K(V_2 - 2V_{th})^2} \qquad \text{Eq B.5}$$

Appendix C: Low pass filter frequency response

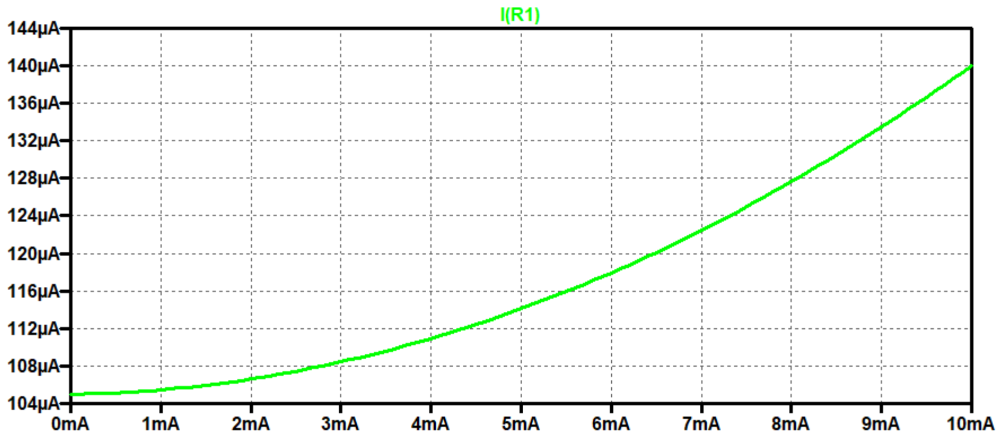


Appendix C.1: Frequency response of the low pass filter to ensure a <1Hz corner frequency

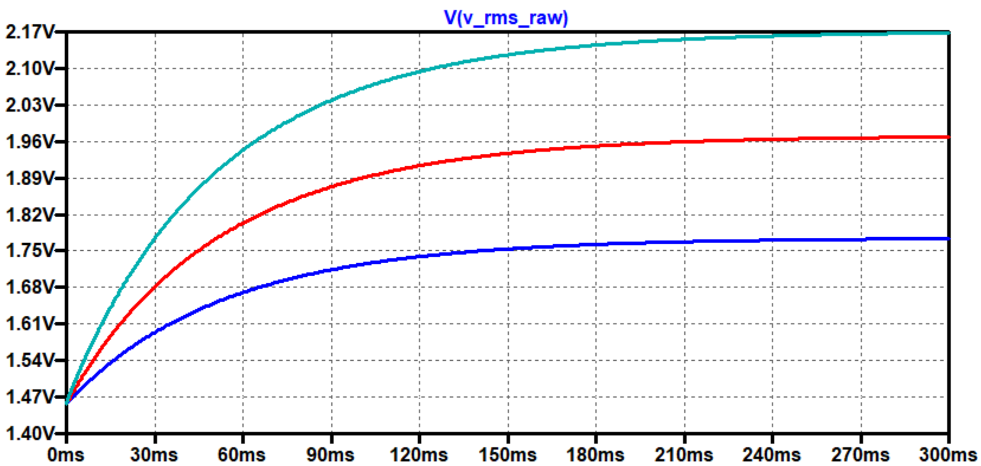
Appendix D: Circuit Validation Waveforms



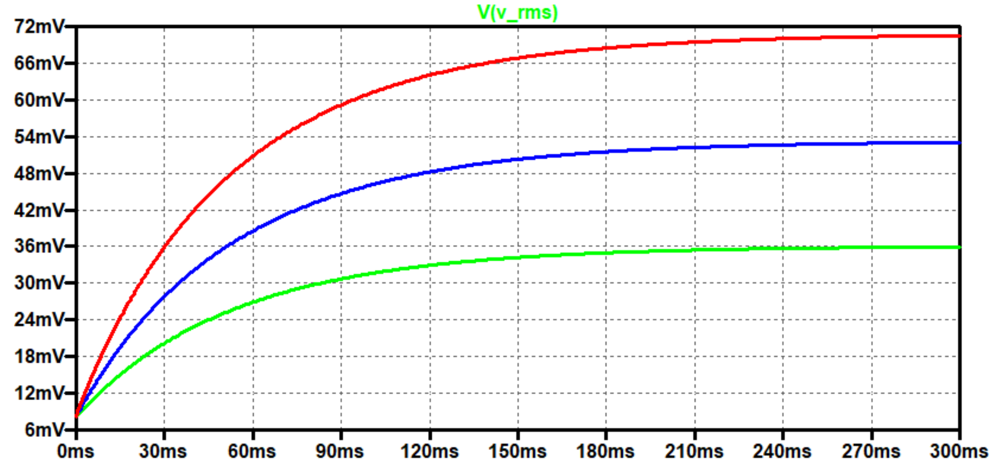
Appendix D.1: Precision Rectifier output waveform for a 10kHz sine wave at 50mV, 75mV and 100mV



Appendix D.2: DC Sweep of output currents from the current square to ensure functionality.



Appendix D.3: Lowpass filtered output of the current squarer divider (Low pass filter was slightly lowered to reduce the simulation time. Thorough testing has been done to ensure these values are accurate for the 470uF capacitor.)



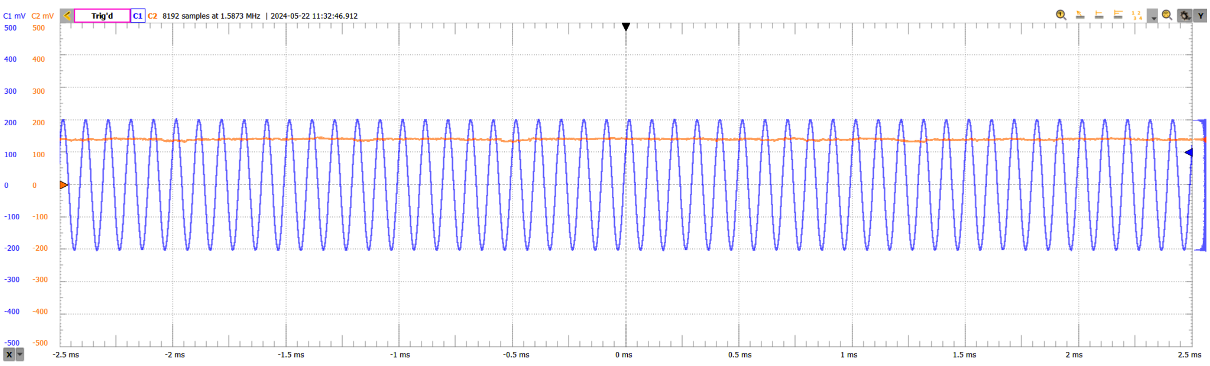
Appendix D.4: Full output waveform for a 10kHz sine wave.

Appendix D.5: Results table for simulated values.

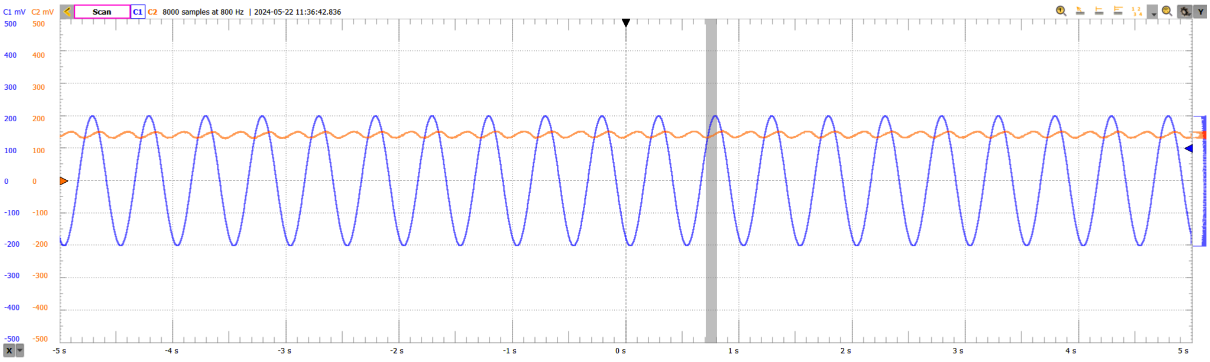
Wave Type	Amplitude (mV)	Frequency (Hz)	Ideal RMS Value (mV)	Measured Value (mV)	% error
DC	50	0	50	51.88	-3.76%
	75	0	75	76.9	-2.53%
	100	0	100	102.14	-2.14%
Sine Wave	50	2	35.35	35.71	-1.02%
	75	2	53.025	52.95	0.14%
	100	2	70.7	70.57	0.18%
	50	10k	35.35	35.92	-1.61%
	75	10k	53.025	53.11	-0.16%
	100	10k	70.7	70.65	0.07%
Full Rectified Sine Wave	50	2	35.35	35.88	-1.50%
	75	2	53.025	53.09	-0.12%
	100	2	70.7	70.51	0.27%
	50	10k	35.35	35.94	-1.67%
	75	10k	53.025	53.21	-0.35%
	100	10k	70.7	70.6	0.14%
Half Rectified Sine Wave	50	2	25	24.1	3.60%
	75	2	37.5	36.2	3.47%
	100	2	50	47.96	4.08%

	50	10k	25	24.4	2.40%
	75	10k	37.5	36.56	2.51%
	100	10k	50	48.22	3.56%
Triangle Wave	50	2	28.86	29.18	-1.08%
	75	2	36.56	42.96	0.79%
	100	2	48.22	57.39	0.60%
	50	10k	28.86	29.2	-1.15%
	75	10k	36.56	42.98	0.74%
	100	10k	48.22	57.41	0.56%
Square Wave	50	2	50	51.86	-3.72%
	75	2	75	76.91	-2.55%
	100	2	100	101.98	-1.98%
	50	10k	50	51.88	-3.76%
	75	10k	75	76.9	-2.53%
	100	10k	100	102.14	-2.14%
PWM Signal (20% duty cycle)	50	2	22.36	19.1	14.58%
	75	2	33.54	29.13	13.15%
	100	2	44.72	38.41	14.11%
	50	10k	22.36	19.42	13.15%
	75	10k	33.54	29.25	12.79%
	100	10k	44.72	38.46	14.00%

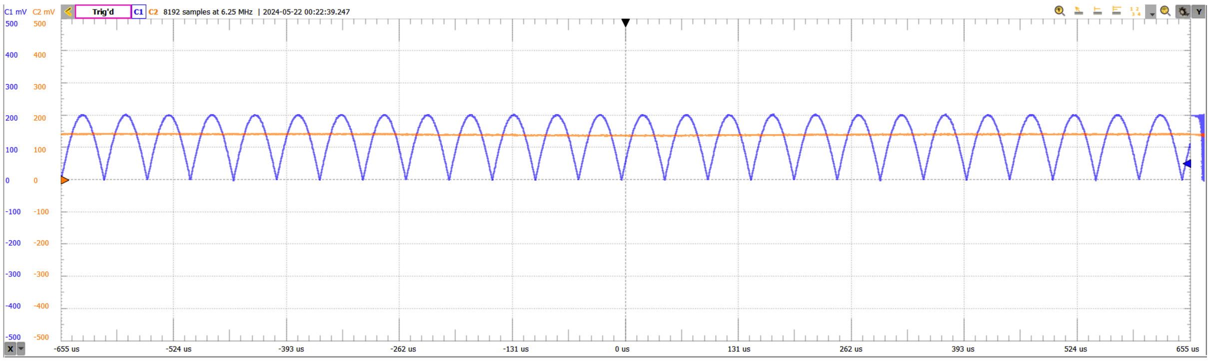
Appendix E: Waveform outputs



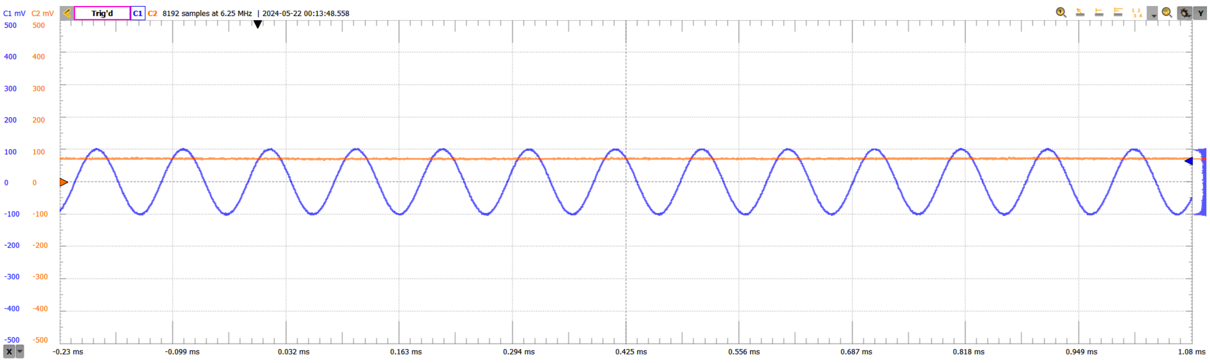
Appendix E.1: Sine input, 200mV, 10kHz



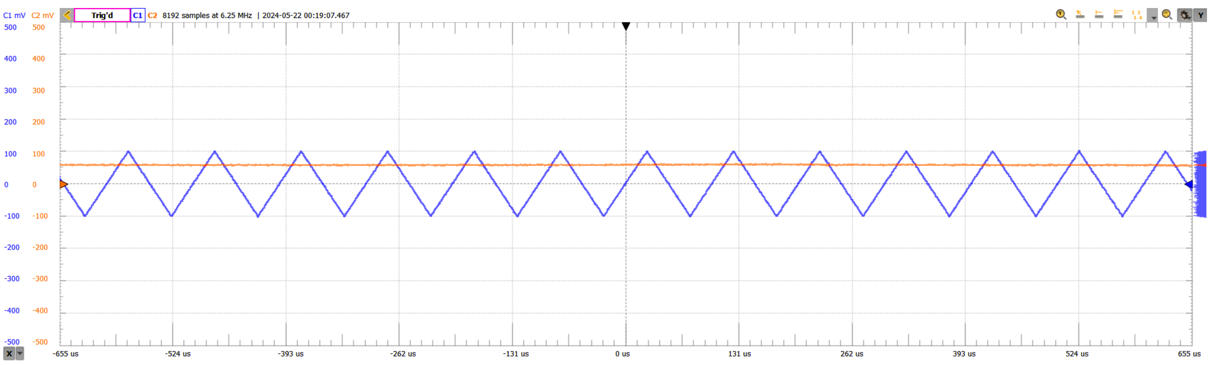
Appendix E.2: Sine input, 200mV, 2Hz



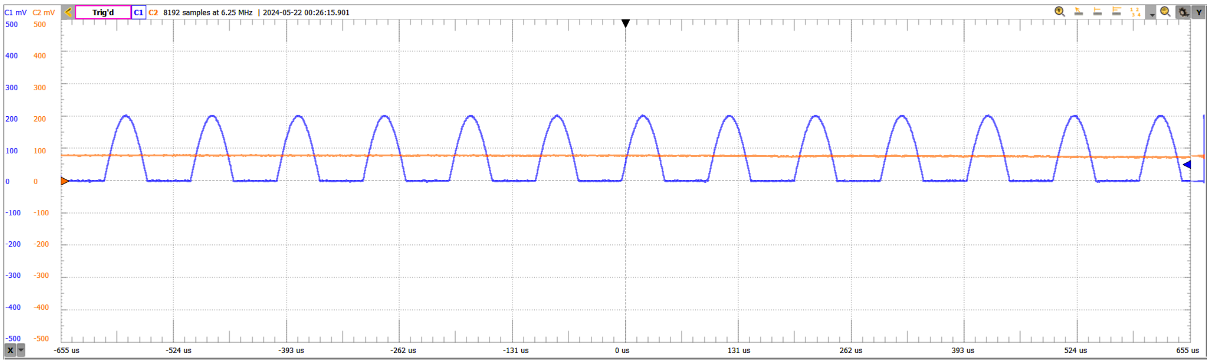
Appendix E.3: Full-wave rectified, 100mV, 10kHz



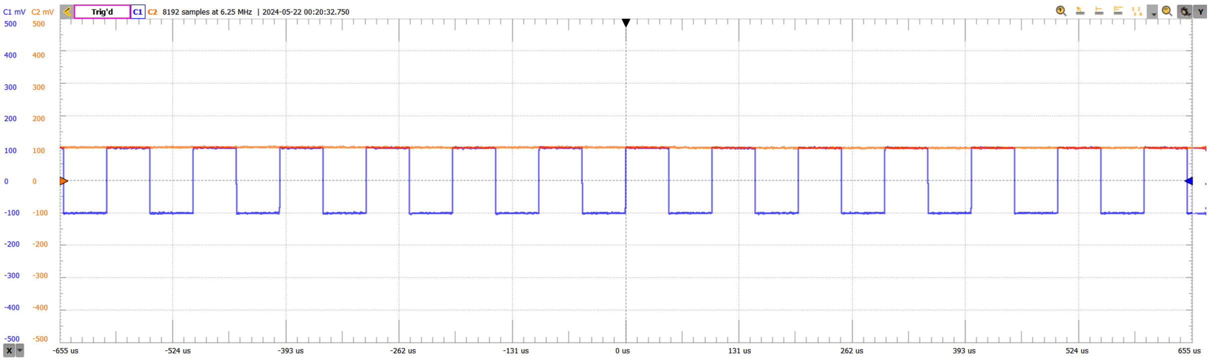
Appendix E.4: Sine input, 100mV, 10kHz



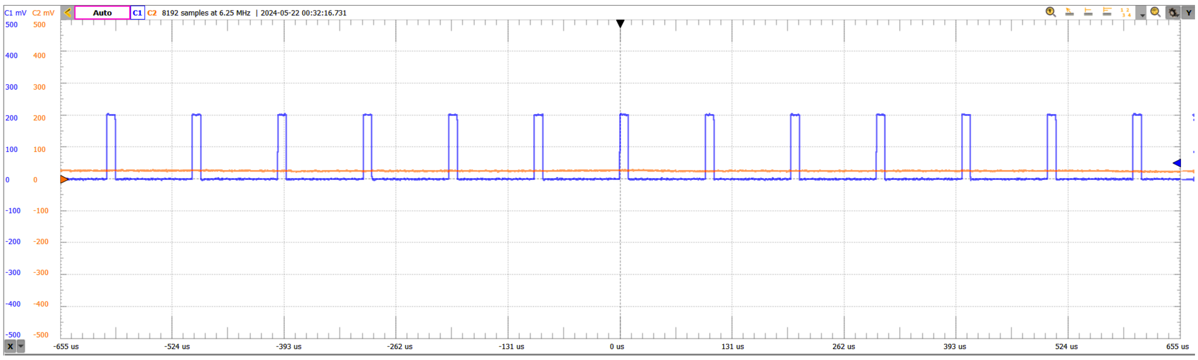
Appendix E.5: Triangle input, 100mV, 10kHz



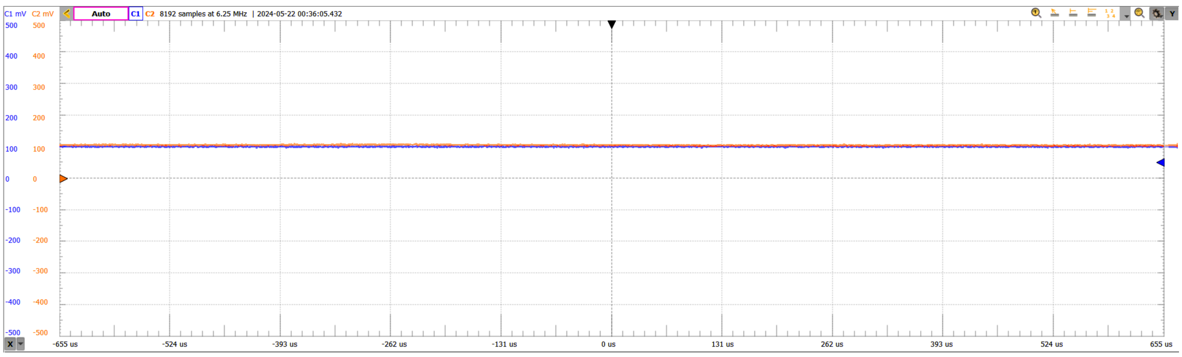
Appendix E.6: Half wave rectified input, 100mV, 10kHz



Appendix E.7: Square wave input, 100mV, 10kHz



Appendix E.8: 10% PWM inputs, 100mV, 10kHz



Appendix E.9: DC input, 100mV

End of Report